

Learn representation, generation, and computation together

REPRESENTATION: token vectors and component extractors

Token 1  → woman; red coat

Token 2  → man; green sweater

Relation R  → giver = token 1; receiver = token 2
object = blue cup

Visual V  → hands and cup in image tokens

Make informative
components ready
earlier to improve
generation

Vary noise to learn
which components
help predict
other components

Choose component inputs and targets;
share attention across tasks;
specialize prediction modules.

DENOISING ALGORITHM

One progress variable per component:

$$\tau(t) = (\tau_1(t), \dots, \tau_K(t))$$

Predict from current component states.
Learn each component's rate of advance.

TRANSFORMER ARCHITECTURE

Conditions

1, 2, R →  →  → V

1, 2, V →  →  → R

Each task also receives its noisy target.

↑ **Jointly train schedules and transformer** ↓